

Step 2: Find Your Weather Station

FileEditRunTools

1SELECT usaf, name, country, state

2FROM `bigquery-public-data.noaa\_gsod.stations`

3WHERE name LIKE '%LA GUARDIA%'

4AND country = 'US'

Query completed

Using on-demand processing quota

Recent<

Query resultsChatView all jobsSave resultsOpen in

7:03 PM

Job informationResultsVisualizationJSONExecution detailsExecution graph

SELECT usaf, name,...

Rowusafnamecountrystate

1725030LA GUARDIA AIRPORTUSNY

6:58 PM

SELECT COUNT(*) E

Step 3: Preview the Ride Data:

SQLlab3Step3 2026-08-23 10:...

RunSaveSchedule

FileEditRunTools

1SELECT starttime, stoptime, tripduration

2FROM `bigquery-public-data.new\_york\_citibike.citibike\_trips`

3LIMIT 10;

This query will process 1.19 GB when run.

Using on-demand processing quota

Recent<

Query resultsChatView all jobs

10:11 AM

Job informationResultsVisualization

SELECT starttime, s...

Rowstarttimestoptimetripduration

1nullnullnull

2nullnullnull

3nullnullnull

4nullnullnull

5nullnullnull

6nullnullnull

7nullnullnull

8nullnullnull

9nullnullnull

10nullnullnull

Step4: Size up the problem

SQLLab3Step4

Run Save Schedule Share

File Edit Run Tools

```
1 SELECT COUNT(*) AS total_trips
2 FROM `bigquery-public-data.new_york_citibike.citibike_trips`;
```

✓ This query will process 0 B when run.

Recent	Query results	Chat	View all jobs	Save results
12:02 PM	Job information	Results	Visualization	JSON
✓ SELECT COUNT(*) ...	Row	total_trips		
	1	58937715		

Question: Why can't we just export this table to a CSV and open it in pandas?

The table has nearly 59 million rows, too large to comfortably load into memory on a local machine. The aggregation needs to happen inside BigQuery first, before the data is small enough to work with in pandas.

Step5:RidesPerDay Run Save Schedule Share

File Edit Run Tools

```

1 SELECT
2   DATE(starttime) AS ride_date,
3   COUNT(*) AS num_rides
4 FROM `bigquery-public-data.new_york_citibike.citibike_trips`
5 WHERE starttime IS NOT NULL
6 GROUP BY ride_date
7 ORDER BY ride_date;

```

✓ This query will process 405.19 MB when run.

Using on-demand processing quota

Recent ◀ Query results Chat View

▼ 12:12 PM

✓ SELECT DATE(starttime) AS ride_date, COUNT(*) AS num_rides FROM `bigquery-public-data.new\_york\_citibike.citibike\_trips` WHERE starttime IS NOT NULL GROUP BY ride_date ORDER BY ride_date;

Row	ride_date	num_rides
1	2013-07-01	16650
2	2013-07-02	22745
3	2013-07-03	21864
4	2013-07-04	22326
5	2013-07-05	21842
6	2013-07-06	20467
7	2013-07-07	20477
8	2013-07-08	21615
9	2013-07-09	26641

Step6: AddAverageTripDuration Run Save Schedule Share

File Edit Run Tools

```

1 SELECT
2   DATE(starttime) AS ride_date,
3   COUNT(*) AS num_rides,
4   AVG(tripduration) / 60 AS avg_duration_min
5 FROM `bigquery-public-data.new_york_citibike.citibike_trips`
6 WHERE starttime IS NOT NULL
7 GROUP BY ride_date
8 ORDER BY ride_date;

```

✓ This query will process 810.37 MB when run.

Using on-demand processing quota

Recent ◀ Query results Chat View all jobs Save results

▼ 12:18 PM

✓ SELECT DATE(starttime) AS ride_date, COUNT(*) AS num_rides, AVG(tripduration) / 60 AS avg_duration_min FROM `bigquery-public-data.new\_york\_citibike.citibike\_trips` WHERE starttime IS NOT NULL GROUP BY ride_date ORDER BY ride_date;

Row	ride_date	num_rides	avg_duration_min
1	2013-07-01	16650	16.30998798798...
2	2013-07-02	22745	15.96882611562...
3	2013-07-03	21864	16.23880808635...
4	2013-07-04	22326	21.21847397652...
5	2013-07-05	21842	18.04044348808...
6	2013-07-06	20467	19.32626178726...
7	2013-07-07	20477	18.63832348488...
8	2013-07-08	21615	21.49216439201...

Step7:OneStationSixYearsOfW...



Save

Schedule

Share

File Edit Run Tools

```
1 SELECT
2   PARSE_DATE('%Y-%m-%d', CONCAT(year, '-', mo, '-', da)) AS obs_date,
3   temp AS temp_f,
4   'max' AS max_temp_f,
5   'min' AS min_temp_f,
6   wdsp AS wind_speed_knots,
7   prcp AS precip_in
8 FROM `bigquery-public-data.noaa_gsod.gsod20*`
9 WHERE stn = '725030'
10 AND _TABLE_SUFFIX BETWEEN '13' AND '18';
```

✓ This query will process 1.38 GB when run.

Using on-demand processing quota

Recent



Query results

Chat

View all jobs

Save results

Open in



12:28 PM

✓ SELECT PARSE_DA...

Job information

Results

Visualization

JSON

Execution details

Execution graph

Row	obs_date	temp_f	max_temp_f	min_temp_f	wind_speed_knots	precip_in
44	2017-03-12	26.6	32.0	19.9	16.2	0.0
45	2017-05-18	88.1	97.0	66.0	10.1	0.0
46	2017-04-29	76.9	88.0	55.0	8.8	0.06
47	2017-01-01	46.4	48.9	44.1	9.9	0.0
48	2017-11-03	66.1	75.0	57.9	7.6	0.0
49	2017-08-14	76.6	84.9	71.1	5.8	0.0
50	2017-03-30	45.9	59.0	41.0	8.7	0.0

Results per page: 50 1 - 50 of 2191 |< < > >|

Step8TheJoin Run Save Schedule Share

File Edit Run Tools

✓ This query will process 2.17 GB when run.

Query results Chat View all jobs Save results Open in

Job information **Results** Visualization JSON Execution details Execution graph

Row	ride_date	num_rides	avg_duration_min	temp_f	max_temp_f	min_temp_f	wind_speed_knots
1	2013-07-01	16650	16.30998798798...	74.8	78.1	73.4	7.8
2	2013-07-02	22745	15.96882611562...	76.1	82.9	73.0	8.0
3	2013-07-03	21864	16.23880808635...	78.5	84.9	73.9	8.8
4	2013-07-04	22326	21.21847397652...	82.0	91.0	73.9	8.6
5	2013-07-05	21842	18.04044348808...	84.4	93.0	75.9	9.0
6	2013-07-06	20467	19.32626178726...	87.0	95.0	77.0	9.3
7	2013-07-07	20477	18.63832348488...	88.8	98.1	79.0	8.8
8	2013-07-08	21615	21.49216439201...	83.7	98.1	75.0	8.9
9	2013-07-09	26641	16.09885452247...	82.6	91.0	75.0	6.0
10	2013-07-10	25732	15.53195437587...	82.0	91.0	75.0	8.7
11	2013-07-11	24417	16.82863442137...	82.3	91.0	75.0	7.6
12	2013-07-12	19006	15.63512750359...	75.4	86.0	72.0	9.1
13	2013-07-13	26119	18.61566101305...	71.4	78.1	68.0	10.5
14	2013-07-14	29287	19.36267115102...	80.2	93.9	69.1	6.0
15	2013-07-15	28069	15.31828232807...	87.2	93.9	72.0	7.8

Results per page: 50 1 – 50 of 1610

Question: The weather CTE alone had 2,191 rows, but the joined table has ~ 1,610 - Where did the other day go?

Daily_weather covers every calendar day across all of 2013-2018, but *daily_rides* only has data from July 1, 2013 (when citibike launched) through May, 31 2018 (where the trip table ends.) An INNER JOIN only keep rows that match on both sides so all the weather day from before JULY 2013 and after May 2018 have no matching ride data, and get dropped entirely. What's left is just the overlap period, landing around 1,610 days.